**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Inputs, Processes and Outputs

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is algorithmic thinking?

2. What does an encryption utility produce?

A) A smaller file
B) A sorted folder
C) Ciphertext
D) Extra storage space

3. What does a compression utility do?

4. A CPU reads an instruction at memory address 120. Correct both claims. Explain why. (a) MAR holds a calculation result. (b) MDR always holds the next instruction's address.

## Today's objectives

Identify inputs, processing and output for a specified problem. Construct an accurate IPO diagram.

## Key vocabulary - Read, cover, recall

**Input**
Data supplied to a program.

**Processing**
Operations or decisions performed on data.

**Output**
Result produced by an algorithm.

**IPO**
Input-Process-Output model.

**Write one key term and its meaning in your own words:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Start with the result** | Every problem asks for a useful result: a total, decision, message or control signal. Identify what the result should be first. |
| **Identify specific inputs** | Inputs are supplied values such as length and width, a sensor reading or a selected option. Name exact values, rather than saying 'some data'. |
| **Describe the processing** | Processing includes calculations, comparisons or decisions. For example: total = unitPrice x quantity, rather than 'the computer works it out'. |
| **Output completes the task** | Output could be a displayed cost or warning message. Check that it follows from the processing and answers the stated question. |

## I Do - One worked example

**A program takes a rectangle's length and width and calculates its area. Identify the input, processing and output. [4 marks]**

1. The result required is the rectangle's area.

2. Inputs are length and width [1], two values supplied to the algorithm [1].

3. Processing multiplies length x width [1].

4. Output displays the area [1]. A separate result follows the calculation.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A snack shop program calculates the total for five identical snacks. Identify input, processing and output. [4 marks]

2. A temperature monitor displays an ice warning when a reading is below 2°C. Explain input, processing and output. [4 marks]

3. A ticket machine accepts an adult fare and a child fare and displays their total. Give four ordered steps. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Write the line that prints the pixel count.**

```
imageWidth = 10
imageHeight = 8
pixels = imageWidth * imageHeight
```

BLACK PEN: predict or complete the code; test, then improve.

## You Do - Independent exam questions

Q2. Define processing within an algorithm. [2 marks]

Q3. A streaming service calculates subscription cost from months purchased and a price per month. Describe the IPO stages. [4 marks]

Q4. Explain why identifying IPO accurately helps when designing a school uniform ordering algorithm. [6 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Output is produced after suitable processing has been performed on required inputs. ☐ **T** ☐ **F**
2. Every problem asks for a useful result: a total, decision, message or control signal ☐ **T** ☐ **F**
3. Inputs are supplied values such as length and width, a sensor reading or a selected option ☐ **T** ☐ **F**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** Developing a clear, ordered and unambiguous method for solving a problem. ☐

**Q2** C - Ciphertext ☐

**Q3** It reduces the storage space required for data, using a suitable lossless or lossy method. ☐

**Q4** (a) The MAR holds the address being accessed. (b) The MDR holds data or an instruction moving to or from memory. An address is different from the data at it. ☐

## Code First - Model

```
imageWidth = 10
imageHeight = 8
pixels = imageWidth * imageHeight
print(pixels)
```

**Output: 80**

## We Do - Check the model

1. Inputs: price per snack [1] and quantity 5 [1]. Process: multiply price x 5 [1]. Output: the total price [1]. ☐

2. Temperature from the sensor is input [1]. The program compares it with 2°C [1]. If below the threshold [1], it outputs an ice warning [1]. ☐

3. Input adult fare [1]. Input child fare [1]. Add them [1]. Display the sum [1]. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** Operations/calculations performed on inputs [1] to obtain results [1]. ☐

**Q3** Inputs are number of months [1] and monthly price [1]. Processing multiplies them [1]. Output displays the cost [1]. ☐

**Q4** Inputs could be item types and quantities [1] and unit prices [1]. Processing multiplies unit prices and quantities [1] and totals the order [1]. Output gives total price/confirmation [1]. Precise IPO helps plan/test the algorithm [1]. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**